# Energy barriers and Carleson budgets for off-critical zeros of the Riemann zeta function

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**Abstract**

This paper develops an effective near-field barrier against off-critical zeros of the Riemann zeta function in the strip $\frac{1}{2} < \Re s < 0.6$. Writing a hypothetical zero as $\rho = \frac{1}{2} + \eta + i\gamma$ with $\eta \in (0, 0.1)$, we compare a universal windowed phase-cost lower bound (a half-plane Blaschke trigger) to a scale-tracked Carleson-box energy budget for a harmonic log-modulus potential attached to a locally neutralized zeta-ratio. This yields an explicit *protection height* $T_{\text{safe}}(\eta)$ such that any such zero must satisfy $|\gamma| > T_{\text{safe}}(\eta)$. The bound is enormous but fully explicit; it also isolates the single height-growth mechanism in the budget (a term proportional to $L \log\langle|\gamma|\rangle$ with $L = 2\eta$), explaining why the conclusion is height-limited rather than uniform in height.

## 1 Introduction

The Riemann zeta function $\zeta(s)$ extends meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$. Its nontrivial zeros encode fine fluctuations of primes, and the Riemann Hypothesis asserts that all nontrivial zeros lie on the critical line $\Re s = \frac{1}{2}$; see [1, 3]. This paper isolates a complementary statement of *effective* (height-limited) type: we derive an explicit height barrier against zeros that lie strictly to the right of the critical line but remain in the near strip

$$\tfrac{1}{2} < \Re s < 0.6.$$

### Main statement (effective; no all-heights claim)

Write a hypothetical off-critical zero as $\rho = \frac{1}{2} + \eta + i\gamma$ with $\eta \in (0, 0.1)$ and set the natural near-field scale and height parameters

$$L \;:=\; 2\eta \in (0, 0.2), \qquad T \;:=\; |\gamma|, \qquad \langle T\rangle \;:=\; \sqrt{1 + T^2}.$$

The main output is a computable *protection height* $T_{\text{safe}}(\eta)$ such that an off-critical zero at depth $\eta$ cannot occur below height $T_{\text{safe}}(\eta)$. An explicit closed-form definition of $T_{\text{safe}}(\eta)$ (in terms of the budget constants) is given in Definition 8.

**Theorem 1** (Effective near-field barrier)**.** *There exists an explicit, computable function $\eta \mapsto T_{\text{safe}}(\eta)$ on $(0, 0.1)$ with the following property: if $\rho = \frac{1}{2} + \eta + i\gamma$ is a zero of $\zeta(s)$ with $\eta \in (0, 0.1)$, then necessarily*

$$|\gamma| \;>\; T_{\text{safe}}(\eta).$$

*Equivalently, $\zeta(s) \neq 0$ for every point of the form $s = \frac{1}{2} + \eta + it$ with $\eta \in (0, 0.1)$ and $|t| \leq T_{\text{safe}}(\eta)$.*

*Remark* 2 (Magnitude of the barrier). The bound is extremely large. For example (with the specific window and constants used in this paper), one obtains $T_{\text{safe}}(0.1) \approx 10^{74}$ and $T_{\text{safe}}(0.01) \gtrsim 10^{8800}$. The point of Theorem 1 is therefore not that the resulting height is small, but that it is *fully explicit* and arises from a clean analytic inequality.

### Why the result is effective: energy cost versus a scale-tracked Carleson budget

The method is an energy comparison with two inputs:

- **Lower bound (quantized phase cost).** An off-critical zero forces a nontrivial amount of boundary phase winding for a normalized zeta-ratio. After localizing at the natural scale $L = 2\eta$ by a compactly supported window centered at height $\gamma$, this produces a universal *minimum* phase-mass cost
$$L_{\text{rec}} \;=\; 4\arctan(2) \;\approx\; 4.428.$$
- **Upper bound (Carleson energy budget).** The same localized phase mass is bounded above by a Carleson-box energy constant for a harmonic log-modulus potential on a box of height comparable to $L$ above the support interval $[\gamma - 2L, \gamma + 2L]$. This Carleson budget naturally splits into:
  - a *prime-layer* contribution depending only on the local scale $L$, controlled unconditionally by explicit prime-tail bounds and harmonic-analysis estimates;
  - a *zero-layer* contribution depending on the height $T$, whose dominant term is proportional to $L \log\langle T\rangle$ and comes from the Poisson balayage of critical-line zeros.

Since the zero-layer term grows with $T$, the conclusion is inherently height-dependent: the argument rules out an off-critical zero at depth $\eta$ only up to the point where the available Carleson budget can match the fixed vortex cost. This is the precise sense in which the barrier is *effective* rather than *uniform in height*.

### Context within the three-paper split

This paper is the "near-field" component of the split project. Paper I provides an unconditional far-field zero-free region ($\Re s \geq 0.6$) by a certified bounded-real (Schur/Pick) argument. The present paper is logically independent of that certification: it supplies a separate, effective barrier mechanism in the remaining strip $\frac{1}{2} < \Re s < 0.6$.

### Roadmap

Section 2 fixes the windowing and phase conventions and defines the Carleson-box energy. Section 3 proves the quantized phase-cost lower bound (the Blaschke trigger), and Section 4 records the CR–Green upper bound by box energy. Section 5 combines these into the barrier inequality and defines $T_{\text{safe}}(\eta)$. Finally, Section 6 discharges the budget decomposition (primes + zeros), completing the effective near-field exclusion.

## 2 Setup: windowing, phase conventions, and Carleson energy

This section fixes notation for the near-field barrier argument. The main objects are: (i) a compactly supported *window* that localizes near the height of a hypothetical zero, (ii) a boundary *phase* whose

distributional derivative records the vortex created by an off-critical zero, and (iii) a *Carleson box energy* that provides an upper budget for that localized phase mass.

## Half-plane, boundary parameterization, and off-critical zeros

We work on the right half-plane

$$\Omega \;:=\; \{\, s \in \mathbb{C} :\; \Re s > \tfrac{1}{2} \,\},$$

with boundary line $\partial\Omega = \{\, \frac{1}{2} + it :\; t \in \mathbb{R} \,\}$. Write a hypothetical off-critical zero as $\rho = \beta + i\gamma$ with $\frac{1}{2} < \beta < 0.6$, and set

$$\eta := \beta - \tfrac{1}{2} \in (0, 0.1), \qquad T := |\gamma|.$$

The natural near-field length scale associated to depth $\eta$ is

$$L := 2\eta, \qquad I_{\gamma,L} := [\gamma - L, \gamma + L].$$

## Flat-top windows

Fix once and for all an even $C^\infty$ flat-top cutoff $\psi : \mathbb{R} \to [0,1]$ such that

$$\psi \equiv 1 \;\text{ on } [-1,1], \qquad \operatorname{supp}\psi \subset [-2,2].$$

For $L > 0$ and $t_0 \in \mathbb{R}$ define the rescaled window and its normalized version by

$$\psi_{L,t_0}(t) := \psi\!\left(\frac{t - t_0}{L}\right), \qquad \varphi_{L,t_0}(t) := \frac{1}{L}\,\psi\!\left(\frac{t - t_0}{L}\right), \qquad m_\psi := \int_{\mathbb{R}} \psi(t)\,dt.$$

Then $\int_{\mathbb{R}} \varphi_{L,t_0}(t)\,dt = m_\psi$, the support satisfies $\operatorname{supp}\varphi_{L,t_0} \subset [t_0 - 2L, t_0 + 2L]$, and $\varphi_{L,t_0} \equiv L^{-1}$ on $[t_0 - L, t_0 + L]$. We will test boundary distributions against $\psi_{L,\gamma}$ to localize at the zero's height.

## Half-plane Blaschke factors and the phase-cost kernel

For $\rho = \frac{1}{2} + \eta + i\gamma \in \Omega$ define the reflected point across $\partial\Omega$ by

$$\rho^* \;:=\; 1 - \overline{\rho} \;=\; \tfrac{1}{2} - \eta + i\gamma,$$

and define the half-plane Blaschke (pole) factor

$$C_\rho(s) \;:=\; \frac{s - \rho^*}{s - \rho}.$$

On the boundary $s = \frac{1}{2} + it$ one has $|C_\rho(\frac{1}{2} + it)| = 1$ and a direct computation gives the distributional identity

$$\frac{d}{dt} \arg C_\rho\!\left(\tfrac{1}{2} + it\right) \;=\; \frac{2\eta}{(t-\gamma)^2 + \eta^2} \;\geq\; 0.$$

This is the basic *vortex kernel*: it is nonnegative, concentrated on the scale $|t - \gamma| \lesssim \eta$, and integrates to $2\pi$. Windowing against $\psi_{2\eta,\gamma}$ yields the explicit lower bound

$$\int_{\mathbb{R}} \psi_{2\eta,\gamma}(t)\, \frac{2\eta}{(t-\gamma)^2 + \eta^2}\, dt \;\geq\; 4\arctan(2) \;=:\; L_{\mathrm{rec}}.$$

### Phase convention and boundary phase mass

Let $\mathcal{J}$ be a meromorphic function on $\Omega$ such that any zero $\rho \in \Omega$ of $\zeta$ produces a pole of $\mathcal{J}$ at $\rho$ (for example, a normalized zeta-ratio). On the boundary $\partial\Omega$ we select a phase $w : \mathbb{R} \to \mathbb{R}$ so that the distribution $-w'$ is a *nonnegative* boundary distribution and the contribution of a pole at $\rho$ contains the kernel from the Blaschke factor $C_\rho$ above. Concretely, one may take (where boundary values exist)

$$w(t) \;:=\; -\arg \mathcal{J}(\tfrac{1}{2} + it),$$

i.e. work with $\mathcal{J}^{-1}$ so that pole contributions enter $-w'$ with a positive sign. All phase identities in this paper are used only after testing against a smooth compactly supported window, so $-w'$ is interpreted distributionally.

### Carleson boxes and energy

Write points in $\Omega$ as $s = \sigma + it$ with $\sigma > \frac{1}{2}$. For an interval $I \subset \mathbb{R}$ and an aperture parameter $\alpha \geq 1$, define the Carleson box above $I$ by

$$Q(\alpha I) \;:=\; I \times (0, \alpha|I|] \;=\; \{\, (\, t, \sigma - \tfrac{1}{2}\,) : \; t \in I, \; 0 < \sigma - \tfrac{1}{2} \leq \alpha|I| \,\}.$$

Let $U$ be a harmonic function on $\Omega$ (in the application, $U$ is a log-modulus potential associated to $\mathcal{J}$ on regions where $\mathcal{J}$ is holomorphic and nonvanishing). Define the associated (scale-invariant) energy measure

$$d\lambda \;:=\; |\nabla U(\sigma, t)|^2 \, (\sigma - \tfrac{1}{2}) \, dt \, d\sigma,$$

and the box energy by $\lambda(Q(\alpha I)) = \iint_{Q(\alpha I)} d\lambda$. The Carleson box-energy ratio at base interval $I$ is

$$\mathcal{C}_{\mathrm{box}}(U; I) \;:=\; \frac{1}{|I|} \, \lambda(Q(\alpha I)).$$

At the near-field scale $L = 2\eta$ we will apply this with $I = I_{\gamma,L}$. Later sections establish an explicit upper bound for $\mathcal{C}_{\mathrm{box}}(U; I_{\gamma,L})$ as a function of $L$ and $T = |\gamma|$, and combine it with the windowed lower bound coming from the Blaschke factor to obtain the effective barrier.

## 3 Phase-cost lower bound (Blaschke trigger)

This section proves the universal lower bound on localized phase mass forced by a single off-critical pole/zero. The estimate is independent of height $T = |\gamma|$ and depends on the depth $\eta$ only through the localization scale $L = 2\eta$.

**Lemma 3** (Blaschke trigger: quantized phase cost)**.** *Let $\rho = \frac{1}{2} + \eta + i\gamma \in \Omega$ with $\eta > 0$ and set $L := 2\eta$. Assume $\mathcal{J}$ is meromorphic on $\Omega$ with a pole at $\rho$ and that its boundary phase is chosen as in Section 2 so that the distribution $-w'$ is nonnegative and contains the pole contribution associated to the half-plane Blaschke factor $C_\rho$. Then one has the windowed lower bound*

$$\int_{\mathbb{R}} \psi_{L,\gamma}(t) \, (-w'(t)) \, dt \;\geq\; L_{\mathrm{rec}}, \qquad L_{\mathrm{rec}} := 4 \arctan(2). \tag{1}$$

*Proof.* On the boundary line $s = \frac{1}{2} + it$, the pole factor $C_\rho$ satisfies

$$\frac{d}{dt} \arg C_\rho(\tfrac{1}{2} + it) \;=\; \frac{2\eta}{(t-\gamma)^2 + \eta^2} \;\geq\; 0$$

in distributions (Section 2). By hypothesis, $-w'$ is a nonnegative distribution that dominates the pole contribution, hence

$$-w'(t) \;\geq\; \frac{2\eta}{(t-\gamma)^2+\eta^2}$$

in the sense of distributions. Testing against the nonnegative window $\psi_{L,\gamma}$ gives

$$\int_{\mathbb{R}} \psi_{L,\gamma}(t)\,(-w'(t))\,dt \;\geq\; \int_{\mathbb{R}} \psi_{L,\gamma}(t)\,\frac{2\eta}{(t-\gamma)^2+\eta^2}\,dt.$$

Since $L = 2\eta$ and $\psi_{2\eta,\gamma} \equiv 1$ on $[\gamma - 2\eta, \gamma + 2\eta]$, we obtain

$$\int_{\mathbb{R}} \psi_{2\eta,\gamma}(t)\,\frac{2\eta}{(t-\gamma)^2+\eta^2}\,dt \;\geq\; \int_{\gamma-2\eta}^{\gamma+2\eta} \frac{2\eta}{(t-\gamma)^2+\eta^2}\,dt.$$

With the change of variables $u = t - \gamma$, the last integral is explicit:

$$\int_{-2\eta}^{2\eta} \frac{2\eta}{u^2+\eta^2}\,du \;=\; 4\arctan(2) \;=\; L_{\mathrm{rec}}.$$

Combining these inequalities yields (1). □

*Remark* 4 (Why the trigger is "quantized"). The constant $L_{\mathrm{rec}} = 4\arctan(2)$ is universal: it comes from integrating a fixed fraction of the Poisson kernel associated to the pole at scale $L = 2\eta$. In particular, it does not decay as $T \to \infty$. This height-independence is what makes an energy comparison possible: the lower bound stays fixed while the available Carleson budget varies with scale and height.

# 4 Carleson energy upper bound (CR–Green / box budget)

This section records the analytic inequality that upper-bounds the same windowed phase mass from Section 3 by a Carleson-box energy. Combined with Lemma 3, it yields the core "cost versus budget" comparison behind the effective barrier.

## CR–Green window inequality

**Lemma 5** (CR–Green bound for windowed phase mass)**.** *Fix an aperture $\alpha \geq 1$. There exists a constant $C(\psi) = C(\alpha, \psi) < \infty$ with the following property. Let $t_0 \in \mathbb{R}$, $L > 0$, and set the support interval*

$$I^* \;:=\; [t_0 - 2L,\, t_0 + 2L].$$

*Assume $F$ is holomorphic and zero-free on the Carleson box $Q(\alpha I^*)$ and write*

$$\log F \;=\; U + iV$$

*there, with $U, V$ harmonic and $V$ a harmonic conjugate of $U$. Define the boundary phase $w(t) := -V(\frac{1}{2} + it)$ on $I^*$ (so that $-w'(t) = \partial_t V(\frac{1}{2} + it)$ in distributions). Then*

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\leq\; C(\psi)\,\Big(\iint_{Q(\alpha I^*)} |\nabla U(\sigma,t)|^2\,(\sigma - \tfrac{1}{2})\,dt\,d\sigma\Big)^{1/2}. \tag{2}$$

*Proof.* Since $F$ is holomorphic and zero-free on $Q(\alpha I^*)$, the branch $\log F$ is holomorphic there. In particular $U$ and $V$ satisfy the Cauchy–Riemann relations on $Q(\alpha I^*)$, and on the boundary line $\Re s = \frac{1}{2}$ we have (in distributions) $\partial_t V = \partial_\sigma U$. Let $\Psi_{L,t_0}$ denote the harmonic (Poisson) extension to $\Omega$ of the boundary function $\psi_{L,t_0}$. With $w(t) = -V(\frac{1}{2} + it)$, we obtain

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;=\; \int_{\mathbb{R}} \psi_{L,t_0}(t)\,\partial_\sigma U(\tfrac{1}{2} + it)\,dt.$$

Green's identity on the box $Q(\alpha I^*)$ (using that both $U$ and $\Psi_{L,t_0}$ are harmonic there, and that $\psi_{L,t_0}$ is supported in $I^*$) converts this boundary pairing into a Dirichlet pairing of the form

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,\partial_\sigma U(\tfrac{1}{2} + it)\,dt \;=\; \iint_{Q(\alpha I^*)} \nabla U \cdot \nabla \Psi_{L,t_0}\,(\sigma - \tfrac{1}{2})\,dt\,d\sigma.$$

Applying Cauchy–Schwarz gives

$$\int_{\mathbb{R}} \psi_{L,t_0}\,(-w') \;\le\; \Big(\iint_{Q(\alpha I^*)} |\nabla U|^2\,(\sigma - \tfrac{1}{2})\Big)^{1/2} \Big(\iint_{Q(\alpha I^*)} |\nabla \Psi_{L,t_0}|^2\,(\sigma - \tfrac{1}{2})\Big)^{1/2}.$$

The second factor is scale-invariant under $(t, \sigma - \frac{1}{2}) \mapsto (t_0 + Lt,\, L(\sigma - \frac{1}{2}))$, hence depends only on $\alpha$ and the fixed window profile $\psi$. Absorb it into $C(\psi)$ to obtain (2). □

### Carleson box formulation

Recall the Carleson ratio from Section 2:

$$\mathcal{C}_{\text{box}}(U; I^*) \;:=\; \frac{1}{|I^*|} \iint_{Q(\alpha I^*)} |\nabla U|^2\,(\sigma - \tfrac{1}{2})\,dt\,d\sigma.$$

Since $|I^*| = 4L$, the bound (2) can be rewritten as

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\le\; C(\psi)\,\sqrt{|I^*|\,\mathcal{C}_{\text{box}}(U; I^*)} \;=\; C(\psi)\,\sqrt{4L\,\mathcal{C}_{\text{box}}(U; I^*)}. \tag{3}$$

In Sections 5 and 6 we estimate $\mathcal{C}_{\text{box}}(U; I^*)$ for the particular $U$ arising from the zeta-ratio normalization, yielding a computable budget as a function of the near-field scale $L = 2\eta$ and height $T = |t_0|$.

## 5 Energy barrier inequality and definition of $T_{\text{safe}}(\eta)$

This section packages the near-field argument into a single inequality: any off-critical pole forces a fixed phase-cost (Section 3), while the available phase mass is bounded above by a Carleson energy budget (Section 4). Solving the resulting inequality for height produces the explicit protection height $T_{\text{safe}}(\eta)$.

### Barrier inequality: cost versus budget

**Proposition 6** (Energy barrier inequality)**.** *Let $\rho = \frac{1}{2} + \eta + i\gamma \in \Omega$ with $\eta \in (0, 0.1)$ and set $L := 2\eta$ and $t_0 := \gamma$. Assume $\mathcal{J}$ has a pole at $\rho$, and let $w$ be the boundary phase chosen so that $-w'$ is nonnegative and contains the pole contribution (Section 2). Let $I^* = [t_0 - 2L, t_0 + 2L]$ be the support interval of $\psi_{L,t_0}$.*

*If the CR–Green estimate of Lemma 5 applies to a holomorphic/zero-free neutralization of $\mathcal{J}$ on $Q(\alpha I^*)$ with harmonic log-modulus $U$, then*

$$L \cdot \mathcal{C}_{\text{box}}(U; I^*) \;\ge\; \frac{L_{\text{rec}}^2}{4\,C(\psi)^2}. \tag{4}$$

*Proof.* Apply Lemma 3 with $L = 2\eta$ and $t_0 = \gamma$ to get the lower bound

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\geq\; L_{\mathrm{rec}}.$$

Apply (3) to the (neutralized) holomorphic/zero-free object on $Q(\alpha I^*)$ to get the upper bound

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\leq\; C(\psi)\,\sqrt{4L\,\mathcal{C}_{\mathrm{box}}(U;I^*)}.$$

Combining and squaring yields $L_{\mathrm{rec}}^2 \leq 4C(\psi)^2\,L\,\mathcal{C}_{\mathrm{box}}(U;I^*)$, which is (4). □

## Budget model and an explicit protection height

To turn (4) into a height exclusion, one needs an explicit upper bound on the box ratio $\mathcal{C}_{\mathrm{box}}(U;I^*)$ in terms of the near-field scale $L$ and height $T := |t_0|$. The guiding structure is a two-term budget:

- a *prime-layer* term depending only on $L$ (height-independent);
- a *zero-layer* term of size $\asymp L\log\langle T\rangle$ coming from the density of critical-line zeros.

**Proposition 7** (Carleson budget upper bound)**.** *There exist explicit constants $K_0, K_1, \kappa > 0$ such that for all $L \in (0, 0.2]$ and all heights $T \geq 0$,*

$$\mathcal{C}_{\mathrm{box}}(U;I^*) \;\leq\; \underbrace{K_0 + K_1\log\Bigl(1+\frac{\kappa}{L}\Bigr)}_{\mathcal{C}_{\mathrm{prime}}(L)} \;+\; \underbrace{(1 + L\log\langle T\rangle)}_{\mathcal{C}_{\mathrm{zeros}}(L,T)}, \tag{5}$$

*where $I^* = [t_0 - 2L, t_0 + 2L]$ with $|t_0| = T$ and $U$ is the harmonic potential of the neutralized object on $Q(\alpha I^*)$.*

The proof of Proposition 7 is given in Section 6.

**Definition 8** (Protection height)**.** Let $\eta \in (0, 0.1)$ and set $L := 2\eta$. Define

$$T_{\mathrm{safe}}(\eta) \;:=\; \sqrt{\exp\left(2\,\frac{\frac{L_{\mathrm{rec}}^2}{4\,C(\psi)^2} \;-\; L(\mathcal{C}_{\mathrm{prime}}(L)+1)}{L^2}\right) \;-\; 1}, \qquad \mathcal{C}_{\mathrm{prime}}(L) := K_0 + K_1\log\Bigl(1+\frac{\kappa}{L}\Bigr), \tag{6}$$

with the convention that $T_{\mathrm{safe}}(\eta) = 0$ if the quantity inside the square root is $\leq 0$.

**Theorem 9** (Effective near-field exclusion from the budget bound)**.** *Assume Proposition 7. If $\rho = \beta + i\gamma$ is a zero of $\zeta(s)$ with $\frac{1}{2} < \beta < 0.6$ and $\eta = \beta - \frac{1}{2}$, then*

$$|\gamma| \;>\; T_{\mathrm{safe}}(\eta),$$

*where $T_{\mathrm{safe}}$ is defined in* (6)*.*

*Proof.* Let $\rho = \frac{1}{2} + \eta + i\gamma$ be such a zero and set $L := 2\eta$ and $T := |\gamma|$. The barrier inequality (4) gives

$$\frac{L_{\mathrm{rec}}^2}{4\,C(\psi)^2} \;\leq\; L\cdot\mathcal{C}_{\mathrm{box}}(U;I^*).$$

Applying the budget bound (5) yields

$$\frac{L_{\mathrm{rec}}^2}{4\,C(\psi)^2} \;\leq\; L\Bigl(\mathcal{C}_{\mathrm{prime}}(L)+1\Bigr) \;+\; L^2\log\langle T\rangle.$$

Rearranging gives a lower bound on $\log\langle T\rangle$, and the definition (6) is exactly the threshold obtained by solving $\langle T\rangle = \exp(\cdots)$ for $T$. □

*Remark* 10 (Why the result is height-limited). The prime-layer term $\mathcal{C}_{\text{prime}}(L)$ is height-independent and grows only logarithmically as $L \downarrow 0$. The obstruction to an all-heights near-field exclusion is the term $L^2 \log\langle T\rangle$: as $T \to \infty$ it eventually dominates at fixed $L$, so the inequality can no longer contradict the existence of a pole. This is precisely why the present method produces a computable *effective* height bound rather than a uniform zero-free strip.

# 6 Discharging the budget bound: primes + zeros

This section justifies Proposition 7 by decomposing the Carleson energy into two contributions:

- a **prime-layer term** controlled unconditionally and depending only on the scale $L$;
- a **zero-layer term** coming from the density of critical-line zeros near height $T$.

Only standard inputs are used: Mertens-type prime bounds and the Riemann–von Mangoldt zero count; see [2, 3].

## A bookkeeping inequality for sums of potentials

In applications, the potential $U$ is built (after local neutralization to ensure harmonicity) as a sum of simpler harmonic potentials. We record a crude but useful estimate.

**Lemma 11** (Energy subadditivity up to constants)**.** *Let $U_1, U_2$ be harmonic on $Q(\alpha I^*)$ and set $U = U_1 + U_2$. Then*

$$\mathcal{C}_{\text{box}}(U; I^*) \;\le\; 2\,\mathcal{C}_{\text{box}}(U_1; I^*) \;+\; 2\,\mathcal{C}_{\text{box}}(U_2; I^*).$$

*Proof.* Use $|\nabla(U_1 + U_2)|^2 \le 2|\nabla U_1|^2 + 2|\nabla U_2|^2$ pointwise and integrate over $Q(\alpha I^*)$. □

## Prime-layer bound (scale-tracked)

The prime-layer term is height-independent and grows only logarithmically as $L \downarrow 0$. We isolate it as a standalone bound.

**Proposition 12** (Prime-layer Carleson bound)**.** *There exist absolute constants $K_0, K_1, \kappa > 0$ such that for every $L \in (0, 0.2]$ and every $t_0 \in \mathbb{R}$, the prime-layer contribution satisfies*

$$\mathcal{C}_{\text{prime}}(L) \;\le\; K_0 \;+\; K_1 \log\Big(1 + \frac{\kappa}{L}\Big),$$

*where $\mathcal{C}_{\text{prime}}(L)$ denotes the Carleson ratio associated to the prime-controlled part of $U$ on $Q(\alpha I^*)$.*

*Proof.* We indicate the standard bookkeeping behind this bound. At scale $L$, the prime-controlled part of the potential $U$ is built from a windowed prime-frequency superposition; after expanding its Dirichlet energy on $Q(\alpha I^*)$, one obtains a diagonal term (a weighted $\ell^2$ sum over primes) plus an off-diagonal term (a discrete Hilbert-form sum over prime pairs).

The diagonal term produces the logarithmic dependence $\log(1 + \kappa/L)$ by elementary summation together with scale invariance of the box and Mertens-type bounds (see [2, Ch. 1]). The off-diagonal term is controlled by the Montgomery–Vaughan Hilbert inequality (see [2, Ch. 9]), giving at most a constant multiple of the diagonal contribution. Collecting these estimates and absorbing prime tails into $K_0$ yields the stated bound. □

### Zero-layer bound (balayage / zero counting)

The height dependence enters through the density of zeros on the critical line.

**Proposition 13** (Zero-layer Carleson bound)**.** *There exists an absolute constant $C > 0$ such that for every $L \in (0, 0.2]$ and every $t_0 \in \mathbb{R}$ (set $T := |t_0|$), the zero-layer contribution satisfies*

$$\mathcal{C}_{\text{zeros}}(L, T) \;\le\; C\Big(1 + L \log\langle T\rangle\Big).$$

*Proof.* On $Q(\alpha I^*)$ one neutralizes the zeros of $\zeta$ (equivalently $\xi$) in a slightly enlarged box by dividing by the corresponding finite product of half-plane Blaschke factors. The removed factors contribute explicitly and positively to the phase derivative $-w'$; the remaining (neutralized) log-modulus is harmonic on the box. Thus the zero-layer bookkeeping reduces to counting how many critical-line zeros lie at heights comparable to $T$ within an interval of length comparable to $L$.

By the Riemann–von Mangoldt formula (see [3, Thm. 9.3]), the number of critical-line ordinates in such an interval satisfies

$$N(T + O(L)) - N(T - O(L)) \;\ll\; 1 + L \log\langle T\rangle.$$

Each such zero contributes a uniformly bounded amount to the Carleson ratio at scale $L$ (depending only on the fixed aperture), because the Dirichlet energy of a single half-plane Blaschke factor over $Q(\alpha I^*)$ is $O(|I^*|) \asymp L$. After dividing by $|I^*| \asymp L$ in the definition of $\mathcal{C}_{\text{box}}$, this yields an $O(1)$ contribution per zero, and hence the stated bound. □

### Proof of Proposition 7

*Proof of Proposition 7.* Decompose the (neutralized) potential $U$ on $Q(\alpha I^*)$ into a prime-controlled part plus a zero-layer part. Applying Lemma 11 and absorbing absolute constants into $K_0, K_1$ and the leading "+1" term gives

$$\mathcal{C}_{\text{box}}(U; I^*) \;\le\; \mathcal{C}_{\text{prime}}(L) \;+\; \mathcal{C}_{\text{zeros}}(L, T).$$

Now apply Proposition 12 and Proposition 13 to obtain (5). □

## Conclusion and limitations (effective status)

The argument in this paper is a deterministic energy comparison. An off-critical zero at depth $\eta$ forces a *quantized* windowed phase cost (the Blaschke trigger, Lemma 3), while the same windowed phase mass is bounded above by a Carleson energy budget via a CR–Green inequality (Lemma 5). This produces the barrier inequality (Proposition 6), and with the explicit two-term budget (Proposition 7) yields an explicit protection height $T_{\text{safe}}(\eta)$ (Definition 8) excluding off-critical zeros below that height (Theorem 9).

**What is proved.** For each fixed $\eta \in (0, 0.1)$ the method yields a fully explicit height threshold $T_{\text{safe}}(\eta)$ such that no zeros of $\zeta(s)$ can occur in

$$\Big\{ s \in \mathbb{C} : \; \tfrac{1}{2} + \eta < \Re s < 0.6, \; |\,\Im s\,| \le T_{\text{safe}}(\eta) \Big\}.$$

All inputs used to define $T_{\text{safe}}$ are classical (prime tail bounds, harmonic analysis on the half-plane, and the Riemann–von Mangoldt formula).

**What is *not* proved.** This paper does *not* prove a zero-free near-field strip uniformly for all heights, and it does *not* prove the Riemann Hypothesis. The conclusion is explicitly height-limited.

**The single obstruction to an all-heights near-field exclusion.** The obstruction is transparent in the budget bound (5): the prime-layer term is height-independent, but the zeros term contains the growth

$$\mathcal{C}_{\text{zeros}}(L,T) \asymp 1 + L\log\langle T\rangle.$$

After inserting $L = 2\eta$ into the barrier inequality, this produces the factor $L^2 \log\langle T\rangle$ that eventually dominates as $T \to \infty$ (Remark 10). Any route to an all-heights near-field result within this framework would therefore require a substantially stronger input controlling the zeros contribution uniformly in $T$.

## Statements and Declarations

**Competing interests.** The author declares no competing interests.

**Data and materials availability.** This paper contains no new computational artifacts beyond the repository materials used in the unconditional far-field certification (Paper I of this series). All files needed to compile this manuscript are included in the repository.

**Reproducibility.** All analytic arguments are intended to be reproducible from the stated lemmas and cited sources. When combined with the certified far-field audit of Paper I, the series provides a fully auditable record of the unconditional $\Re s \geq 0.6$ step and the effective near-field barrier mechanism.

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